LIBRARY BOOK MANAGEMENT SYSTEM

1. Project Title : Library Book Management System Using Java

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Git Hub : https://github.com/black-hand07/LibraryBook_Management_System.git

2. Project Overview :

The Library Book Management System is a console-based Java application developed to manage book records in a library. The system allows users to perform CRUD (Create, Read, Update, Delete) operations on book records and stores the data permanently using file handling .

The application uses :

- Java Collections Framework (ArrayList)
- File Handling (File, FileWriter , Scanner)
- Exception Handling (try-catch)
- Menu-driven user interface (while loop)

The system maintains information such as Book ID, Book Name, Author Name, Category, Publication Year, and Book Status.

3. Objective

The main objective of this project is to:

- Manage library books efficiently.
- Store and retrieve book records.
- Implement CRUD operations.
- Demonstrate Java concepts such as:
  - ➢ Classes and Objects
  - ➢ ArrayList
  - ➢ File Handling
  - ➢ Exception Handling
  - ➢ Loops and Conditional Statements

4. Technologies Used

- ✓ Java - Programming Language
- ✓ ArrayList - Dynamic Data Storage
- ✓ File Handling - Permanent Data Storage
- ✓ Scanner Class - User Input
- ✓ Exception Handling - Error Management
- ✓ Text File (Book.txt) - Database Alternative

5. Attributes (Methods)

- **Add Book**

  The user can add a new book by entering:
  - Book ID
  - Book Name
  - Author Name
  - Category
  - Published Year
  - Status

    Validation : Duplicate book ID's are not allowed

- **View Book**

  Displays all available books stored in the system.
  Information Displayed :
  - Book ID
  - Book Name
  - Author Name
  - Category
  - Published Year
  - Status

- **Search Book**

  Allows the user to search a book using Book ID.
  Output :
  - Complete details of the book if found.
  - "Book not found" message if unavailable.

❖ **Update Book**

Allow Modification of :

- ➢ Book name
- ➢ Author Name
- ➢ Category
- ➢ Published Year
- ➢ Status

The user enters Book ID and updates the record.

❖ **Delete Book**

Removes a book record permanently from**:**

- ➢ ArrayList
- ➢ Book.txt file

❖ **Exit**

Terminates the Application

6. Program Structure

```
Book
|
├── Attributes
|   ├── bookId
|   ├── bookName
|   ├── authorName
|   ├── category
|   ├── publicationYear
|   └── status
|
├── Methods
|   ├── addBook()
|   ├── viewBooks()
|   ├── searchBook()
|   ├── updateBook()
|   ├── deleteBook()
|   ├── saveToFile()
|   ├── loadFromFile()
|   └── main()
```

7. Program Flow

```
Start
  |
  ▼
Load books.txt
  |
  ▼
Display Menu
  |
  ├── Add Book
  |       |
  |       ▼
  |   Save File
  |
  ├── View Books
  |
  ├── Search Book
  |
  ├── Update Book
  |       |
  |       ▼
  |   Save File
  |
  ├── Delete Book
  |       |
  |       ▼
  |   Save File
  |
  └── Exit
        |
        ▼
       End
```

8. Sample Code

```java
static void addBook() {
    System.out.print("Enter Book ID : ");
    int id = sc.nextInt();
    sc.nextLine();
    for (Book b : books) {
        if (b.bookId == id) {
            System.out.println("Book ID already exists.");
            return;
        }
    }
    System.out.print("Enter Book Name : ");
    String name = sc.nextLine();
    System.out.print("Enter Author Name : ");
    String author = sc.nextLine();
    System.out.print("Enter Category : ");
    String category = sc.nextLine();
    System.out.print("Enter Publication Year : ");
    int year = sc.nextInt();
    sc.nextLine();
    System.out.print("Enter Status (Available/Issued) : ");
    String status = sc.nextLine();
    books.add(new Book(id, name, author, category, year, status));
    saveToFile();
    System.out.println("Book added successfully.");
}
```

```java
static void viewBooks() {

    if (books.isEmpty()) {
        System.out.println("No books available.");
        return;
    }

    for (Book b : books) {

        System.out.println(
                b.bookId + " | " +
                b.bookName + " | " +
                b.authorName + " | " +
                b.category + " | " +
                b.publicationYear + " | " +
                b.status);
    }
}
```

```java
static void searchBook() {

    System.out.print("Enter Book ID : ");
    int id = sc.nextInt();

    for (Book b : books) {

        if (b.bookId == id) {

            System.out.println("Book Found");
            System.out.println("Name : " + b.bookName);
            System.out.println("Author : " + b.authorName);
            System.out.println("Category : " + b.category);
            System.out.println("Year : " + b.publicationYear);
            System.out.println("Status : " + b.status);
            return;
        }
    }

    System.out.println("Book not found.");
}
```

```java
static void deleteBook() {

    System.out.print("Enter Book ID : ");
    int id = sc.nextInt();

    for (Book b : books) {

        if (b.bookId == id) {

            books.remove(b);

            saveToFile();

            System.out.println("Book deleted successfully.");
            return;
        }
    }

    System.out.println("Book not found.");
}
```

9. Output Screenshots

```
******** LIBRARY BOOK MANAGEMENT SYSTEM ********
1. Add Book
2. View Books
3. Search Book
4. Update Book
5. Delete Book
6. Exit
************************************************
Enter Choice : 1
************************************************
Enter Book ID : 22
Enter Book Name : The Cure At Troy
Enter Author Name : Sophocles' philoctetes
Enter Category : poetry
Enter Publication Year : 2018
Enter Status (Available/Issued) : Available
Book added successfully.
```

```
******** LIBRARY BOOK MANAGEMENT SYSTEM ********
1. Add Book
2. View Books
3. Search Book
4. Update Book
5. Delete Book
6. Exit
************************************************
Enter Choice : 2
************************************************
21 | Dracula | Bramstoker | horror | 2018 | Yes
22 | The Cure At Troy | Sophocles' philoctetes | poetry | 2018 | Available
23 | The Greek Plays | sophocles | Drama | 2016 | Available
```

```
******** LIBRARY BOOK MANAGEMENT SYSTEM ********
1. Add Book
2. View Books
3. Search Book
4. Update Book
5. Delete Book
6. Exit
************************************************
Enter Choice : 3
************************************************
Enter Book ID : 21
Book Found
Name : Dracula
Author : Bramstoker
Category : horror
Year : 2018
Status : Yes
```

```
******** LIBRARY BOOK MANAGEMENT SYSTEM ********
1. Add Book
2. View Books
3. Search Book
4. Update Book
5. Delete Book
6. Exit
************************************************
Enter Choice : 4
************************************************
Enter Book ID : 21
Enter New Book Name : Lucky Wolf
Enter New Author Name : R.N.Rovoleh
Enter New Category : Drama
Enter New Publication Year : 2021
Enter New Status : Available
Book updated successfully.
```

```
******** LIBRARY BOOK MANAGEMENT SYSTEM ********
1. Add Book
2. View Books
3. Search Book
4. Update Book
5. Delete Book
6. Exit
************************************************
Enter Choice : 5
************************************************
Enter Book ID : 21
Book deleted successfully.
```

10.Advantages

- Simple and User Friendly
- No database required.
- Permanent data storage using files.
- Easy maintenance.
- Demonstrates core Java concepts.

11. Limitations

- Console-based interface.
- Single-user application.
- No authentication mechanism.
- Data stored in text files only.

12.Future Enhancement

- Graphical User Interface (GUI)
- Database Integration (MySQL)
- User Authentication
- Book Issue and Return Module
- Report Generation
- Search Filters

13.Conclusion

The Library Book Management System successfully manages book records using Java, ArrayList, File Handling, and Exception Handling. The project demonstrates the implementation of CRUD operations and data persistence without relying on external databases. It provides a practical understanding of Java programming concepts and serves as an effective internship-level project.